IV. Experimental Results and Analysis

We evaluate the proposed method through three research questions (RQs) concerning known-class recognition, unknown-class rejection, and component contributions:

- **RQ1: Does the proposed method offer advantages in known-class recognition?**
- **RQ2: Can the proposed method effectively identify and reject unknown services and attacks?**
- **RQ3: Do the individual views and the empirical support mechanism contribute to rejection performance?**

We design corresponding experiments to address these questions. For RQ1, Section 4.2 compares known-class recognition against existing methods on tasks derived from ISCX-VPN, ISCX-Tor, and other datasets at different class granularities. For RQ2, Section 4.3 evaluates rejection of unknown services and attacks through complete-service holdout on VNAT, class holdout on USTC-TFC2016, and unknown-attack detection on CIC-IDS2017. Section 4.5 further examines the effects of the rejection threshold and changes in the proportion and composition of unknown classes. For RQ3, Section 4.4 compares rejection scores on identical frozen representations and conducts view ablations to assess the contributions of local empirical support and individual view branches.

A. Experimental Setup

1) Datasets and Task Splits: We use bidirectional flows as samples and evaluate on ISCX-VPN, ISCX-Tor, USTC-TFC2016, CIC-IDS2017, and VNAT.

ISCX-VPN and ISCX-Tor retain their six and seven original service labels, respectively, for closed-set comparisons with existing methods. We additionally merge Chat, Email, and VoIP into Communication to construct coarse-grained service tasks with four and five classes. Both task groups use identical flows and role assignments, while the adapters and classification heads train on their respective labels. USTC-TFC2016 uses three class-holdout configurations, A-1, A-2, and A-3, with known/unknown class counts of 19/1, 17/3, and 15/5, respectively. Within each original source split, we deterministically sample 10% of the flows from each class. A-1 holds out Tinba; A-2 holds out Geodo, Htbot, and Tinba; A-3 additionally holds out Miuref and Neris relative to A-2.

CIC-IDS2017 supports two separate experiments. The known-class classification task includes BENIGN, DoS Slowhttptest, and PortScan. For unknown-attack diagnosis, only BENIGN and PortScan serve as known classes, and the corresponding model evaluates ten unknown attacks.

VNAT maps ten applications to four service classes: Communication, File Transfer, Remote Access, and Streaming. Communication includes Skype and Zoiper; File Transfer includes rsync, scp, and sftp; Remote Access includes SSH and RDP; Streaming includes Netflix, Vimeo, and YouTube.

VNAT evaluation operates at the service-class level. Each fold retains the application composition and capture-group coverage of the frozen split.

3) Comparison Methods: We compare representative encrypted-traffic recognition methods covering closed-set classification and unknown-class detection.

(1) Methods for known-class classification:

- **TrafficFormer:** A Transformer-based method that learns contextual network-traffic features through attention for encrypted-traffic classification.
- **YaTC:** A method that learns general traffic representations through self-supervised pretraining and identifies traffic classes through downstream fine-tuning.
- **Open-Detect:** A method based on a variational autoencoder and Gaussian class prototypes that promotes intraclass compactness and interclass separability through generative and discriminative constraints.
- **RoNeTC:** A deep learning method for encrypted-traffic classification that learns traffic features for class discrimination.
- **TFE-GNN:** A method that represents network traffic as graphs and uses a graph neural network to extract interaction features for classification.
- **Trident:** A classification method based on traffic statistics that uses behavioral information for fine-grained class recognition.

(2) Methods for unknown-class detection:

- **Open-Detect Native:** A method that learns latent traffic probability distributions with a variational encoder and identifies unknown classes using the minimum KL divergence between a sample's posterior distribution and Gaussian class prototypes.
- **MSP:** A confidence measure based on the maximum softmax probability of the classifier, which identifies low-confidence samples as unknown.
- **Energy:** An energy score computed from classifier logits that rejects samples based on score differences between known and unknown samples.
- **DES-v0:** A distance score based on empirical centroids of known classes that measures a test sample's deviation from known-class feature distributions.
- **DES-v1:** An extension of empirical centroids that incorporates within-class nearest-neighbor support to assess a sample's local feature consistency with known classes.

MSP, Energy, DES-v0, and DES-v1 use the same frozen fused features and classification outputs. Each score independently calibrates its rejection threshold on the known-class validation set to ensure a fair comparison of rejection strategies.

4) Evaluation Metrics and Statistical Conventions: We evaluate known-class classification using Accuracy, Macro-F1, and Weighted-F1. Macro-F1 assigns equal weight to each class, whereas Weighted-F1 weights classes by their sample counts. For unknown-class detection, we report AUROC, AUPRC, unknown-class F1, the unknown false acceptance rate (UFAR), and the known-class false rejection rate (Known FRR). Unknown samples constitute the positive class. AUROC and AUPRC measure overall unknown-detection ranking performance and precision–recall performance, respectively. Unknown-class F1 measures detection performance at a fixed threshold.

Let D_U denote the unknown test set, D_K^{test} the known test set, $s(x)$ the unknown-detection score, and τ the rejection

TABLE 4-1
CLASS AND ID MAPPINGS FOR USTC-TFC2016, ISCX-VPN, AND ISCX-TOR

ID	Class	ID	Class	ID	Class	ID	Class	ID	Class
U0	Gmail	U7	SMB	U14	MySQL	V0	Chat	T0	Chat
U1	FTP	U8	BitTorrent	U15	Htbot	V1	Email	T1	Email
U2	Nsis-ay	U9	WorldOfWarcraft	U16	Tinba	V2	File Transfer	T2	File Transfer
U3	Facetime	U10	Shifu	U17	Skype	V3	P2P	T3	P2P
U4	Weibo	U11	Outlook	U18	Miuref	V4	Streaming	T4	Audio-streaming
U5	Cridex	U12	Virut	U19	Neris	V5	VoIP	T5	Video-streaming
U6	Zeus	U13	Geodo	—	—	—	—	T6	VoIP

Note: U denotes USTC-TFC2016, V denotes ISCX-VPN, and T denotes ISCX-Tor.

TABLE 4-2
EXPERIMENTAL SCENARIOS AND KNOWN/UNKNOWN CLASS SPLITS

Scenario	Known Classes (KC)	Unknown Classes (UC)
A-1	USTC-TFC2016 \ {U16}	U16 (Tinba)
A-2	USTC-TFC2016 \ {U13, U15, U16}	U13 (Geodo), U15 (Htbot), U16 (Tinba)
A-3	USTC-TFC2016 \ {U13, U15, U16, U18, U19}	U13 (Geodo), U15 (Htbot), U16 (Tinba), U18 (Miuref), U19 (Neris)
VPN-4C	{V0+V1+V5, V2, V3, V4}	None (closed-set)
Tor-5C	{T0+T1+T6, T2, T3, T4, T5}	None (closed-set)

TABLE 4-3
SAMPLE COMPOSITION FOR VNAT COMPLETE-SERVICE HOLDOUT

Held-out unknown service	Known flows	Unknown flows
Communication	21,241	2,208
File Transfer	17,579	5,870
Remote Access	9,842	13,607
Streaming	21,685	1,764

threshold. Then,

$$\text{UFAR} = \frac{|\{x \in D_U : s(x) \le \tau\}|}{|D_U|},$$

$$\text{Known FRR} = \frac{|\{x \in D_K^{test} : s(x) > \tau\}|}{|D_K^{test}|}.$$

UFAR is the proportion of unknown flows incorrectly accepted as known, and Known FRR is the proportion of known flows incorrectly rejected. The primary rejection threshold is the 95th percentile (P95) of detection scores on the known-class validation set, computed using the `higher` quantile rule. Throughout, a sample is unknown if $s(x) > \tau$. Threshold-sensitivity experiments additionally consider P90 and P99.

Some paired experiments construct known/unknown test subsets with a 1:1 ratio using flow identifiers fixed in advance to examine the effect of test-sample proportions. We compute AUPRC as average precision.

B. Known-Class Recognition Performance

Known-class classification and unknown-class rejection address class discrimination and unknown detection, respectively, in open-set recognition. They are separate tasks with different evaluation objectives. To assess classification of different types of encrypted traffic, we first conduct closed-set experiments using the original service labels of four datasets and compare against six representative methods.

Compared with YaTC, the strongest baseline, our method improves Macro-F1 by 3.37 and 4.12 percentage points on VPN and Tor, respectively. This improvement primarily stems from differences in feature representation. YaTC mainly extracts traffic features from multilevel byte representations. Our method jointly models sequence dependencies, flow interactions, and byte structure, integrating complementary views through Gaussian adaptation and feature fusion to improve discrimination between service classes. Section 4.4.2 examines the contributions of individual branches to unknown rejection through controlled ablation.

We further assess applicability across label granularities and task types using the four-class VPN coarse-service task, five-class Tor coarse-service task, 17-class USTC-TFC2016 known-class task, and three-class balanced CIC-IDS2017 classification task. Table 4-6 summarizes the results.

Our method achieves Macro-F1 scores of 98.76% and 99.75% on USTC and CIC, respectively, and 91.94% and 89.94% on the coarse-service VPN and Tor tasks. On CIC, class-level F1 scores for BENIGN, Slowhttptest, and PortScan reach 99.84%, 99.62%, and 99.78%, respectively, indicating relatively consistent recognition across classes. The 98.76% Macro-F1 across 17 known USTC classes indicates that the three-view representation can discriminate feature differences among a larger number of classes.

C. Unknown-Class Rejection Performance

Closed-set classification performance alone does not fully characterize rejection of unknown traffic. We therefore conduct open-set detection experiments on VNAT, USTC-TFC2016, and CIC-IDS2017 to assess unknown-class detection under different class-holdout settings.

Overall rejection performance. We compare our method with Open-Detect on VNAT under complete-service holdout. Table 4-7 reports AUROC, unknown-class F1, UFAR, and Known FRR for four unknown-service scenarios.

Table 4-7 shows that our method outperforms Open-Detect in AUROC and unknown-class F1 in all four service-holdout scenarios. For File Transfer, AUROC and unknown-class F1

TABLE 4-5
CLOSED-SET CLASSIFICATION PERFORMANCE WITH ORIGINAL SERVICE LABELS

Task	**ISCX-VPN**			**ISCX-Tor**		
Method	Accuracy ↑	Macro-F1 ↑	Weighted-F1 ↑	Accuracy ↑	Macro-F1 ↑	Weighted-F1 ↑
TrafficFormer	84.91	85.66	84.21	82.54	81.49	82.73
YaTC	85.64	87.55	86.39	84.25	82.66	84.21
Open-Detect	71.45	73.01	71.19	54.52	47.69	52.38
RoNeTC	69.99	70.76	69.62	59.36	51.42	57.60
TFE-GNN	62.95	64.39	62.82	59.98	53.98	57.94
Trident	61.11	63.16	60.60	63.02	60.34	62.91
Our method	**89.21**	**89.92**	**88.96**	**88.28**	**86.78**	**88.22**

TABLE 4-6
KNOWN-CLASS CLASSIFICATION RESULTS OF THE CURRENT THREE-VIEW METHOD (%)

Dataset and task	Classes	Train/validation/test flows	Accuracy ↑	Macro-F1 ↑	Weighted-F1 ↑
ISCX-VPN, service classification	4	8,764/1,098/1,093	91.58	91.94	91.45
ISCX-Tor, service classification	5	8,946/1,118/1,117	89.62	89.94	89.67
USTC-TFC2016, A-2 known classification	17	34,665/4,333/4,333	98.43	98.76	98.43
CIC-IDS2017, three-class balanced task	3	35,516/3,912/2,536	99.80	99.75	99.80

TABLE 4-7
COMPARISON OF UNKNOWN-CLASS DETECTION PERFORMANCE ON VNAT UNDER COMPLETE-SERVICE HOLDOUT

Metric	**AUROC ↑**		**Unknown F1 ↑**		**UFAR ↓**		**Known FRR ↓**	
Service scenario	Open-Detect	DES-v1	Open-Detect	DES-v1	Open-Detect	DES-v1	Open-Detect	DES-v1
Communication	81.8	**99.5**	34.2	**98.5**	78.9	**0.1**	**3.5**	4.9
File Transfer	60.0	**99.5**	0.3	**99.3**	99.9	**0.6**	10.0	**4.6**
Remote Access	58.9	**98.8**	5.7	**98.0**	97.0	**3.8**	**2.4**	2.9
Streaming	96.1	**99.1**	90.6	**97.8**	13.0	**0.0**	6.0	**5.5**

increase from 59.97% and 0.30% to 99.48% and 99.34%, respectively. For Remote Access, the corresponding scores increase from 58.88% and 5.74% to 98.76% and 97.98%. The four-fold mean UFAR decreases from 72.19% to 1.09%, and Known FRR decreases from 5.46% to 4.45%. Although Known FRR increases slightly for Communication and Remote Access, the overall false rejection rate decreases, indicating improved unknown-service rejection.

To exclude the influence of known/unknown sample proportions in the test set, we additionally evaluate on predefined paired 1:1 subsets. Our method achieves mean AUROC, AUPRC, and unknown-class F1 scores of 99.25%, 98.93%, and 97.32%, respectively, exceeding Open-Detect's 74.17%, 68.24%, and 32.02%. These results indicate that differences in test-set class proportions alone do not explain the detection advantage.

Rejection under different unknown-class configurations. We use the A-1, A-2, and A-3 class-holdout scenarios on USTC-TFC2016 to assess changes in unknown-class counts and composition. Fig. 4-1 presents the results.

Our method achieves higher unknown-class F1 and lower UFAR in both A-2 and A-3. In A-2, unknown-class F1 increases from 50.38% to 67.41%, and UFAR decreases from 64.34% to 46.06%. In A-3, unknown-class F1 increases from 91.91% to 92.31%, and UFAR decreases from 11.25% to 9.21%. In A-1, both methods achieve zero UFAR and perform similarly. Overall, our method provides effective rejection across these unknown-class configurations, with a more pronounced advantage over Open-Detect in scenarios containing multiple unknown classes.

Fig. 4-2. Recognition results for ten unknown CIC attacks (%, original test-pool proportions, P95).

Unknown attack	Unknown flows	MSP AUROC	DES-v1 AUROC	DES-v1 AUPRC	DES-v1 UFAR ↓
DoS slowloris	5,709	81.47	**99.31**	99.67	0.04
DoS GoldenEye	7,441	33.17	**97.00**	97.72	0.24
DoS Hulk	155,168	9.38	**97.38**	99.88	0.11
FTP-Patator	3,985	98.75	**99.83**	99.86	0.00
SSH-Patator	2,987	86.97	**99.13**	98.93	0.00
Web Attack – Brute Force	1,364	91.61	**99.57**	99.14	0.00
Web Attack – XSS	629	97.80	**99.75**	98.84	0.00
DoS Slowhttptest	5,096	85.97	**99.34**	99.63	0.14
DDoS	76,613	5.82	**94.90**	99.45	16.61
Bot	1,228	62.91	**94.64**	87.58	50.90

Class-level responses to unknown attacks. The CIC-IDS2017 experiment fixes the model trained on BENIGN and PortScan, the training support bank, the P95 threshold, and 2,272 known test flows, then introduces each of ten unknown attacks separately. Table 4-7 reports AUROC for MSP and DES-v1, together with AUPRC and UFAR for DES-v1, to compare conventional classification confidence and empirical support scores across attacks.

DES-v1 achieves UFAR no greater than 0.24% for eight of the ten unknown attacks, indicating effective rejection overall. Known FRR is 7.13% for every attack category. For Hulk, MSP achieves only 9.38% AUROC, whereas DES-v1 reaches 97.38%, indicating that empirical support distance can compensate for classification-confidence failures on some unknown attacks. Bot and DDoS remain challenging, with UFAR values of 50.90% and 16.61%, respectively. This variation indicates that feature overlap between attack types and known classes affects rejection performance.

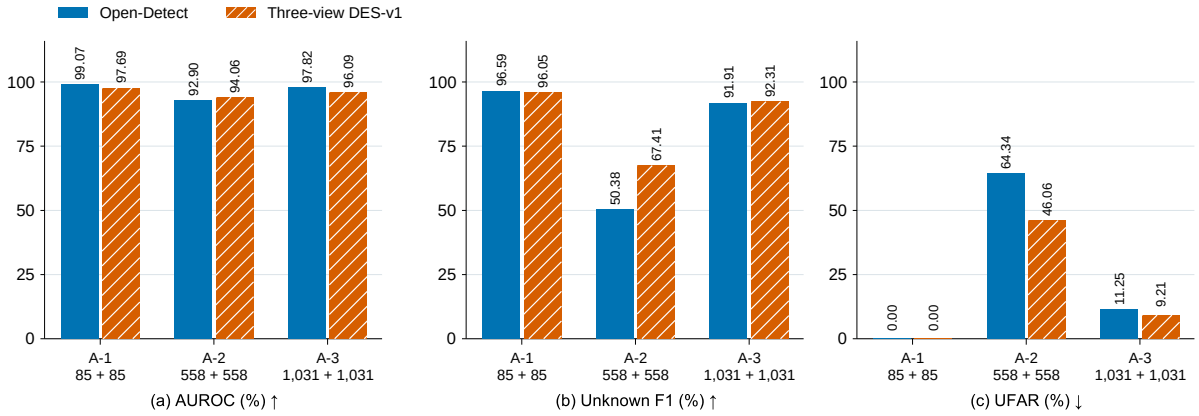


Fig. 4-1. Unknown-class detection performance under three USTC class-holdout configurations. (a) AUROC; (b) unknown-class F1; (c) UFAR.

TABLE 4-8
REJECTION SCORES ON IDENTICAL FROZEN REPRESENTATIONS (%, EQUAL-WEIGHT MEAN OVER FOUR FOLDS, ORIGINAL TEST-POOL PROPORTIONS, P95)

Rejection score	AUROC	AUPRC	Unknown F1	UFAR	Known FRR
MSP	91.83	96.66	90.55	14.63	4.93
Energy	91.73	96.58	89.96	15.40	4.87
DES-v0	97.43	98.79	96.10	5.59	4.87
DES-v1	**99.22**	**99.50**	**98.40**	**1.09**	**4.45**

Note: DES-v0 uses empirical centroids only; DES-v1 combines global and local support. Higher AUROC, AUPRC, and Unknown F1 are better; lower UFAR and Known FRR are better.

TABLE 4-9
VNAT VIEW ABLATIONS UNDER THE SAME CONFIGURATION (%, EQUAL-WEIGHT MEAN OVER FOUR FOLDS, DES-V1, ORIGINAL TEST-POOL PROPORTIONS, P95)

Configuration	Known Macro-F1	AUROC	AUPRC	Unknown F1	UFAR	Known FRR
Full three-view	99.87	99.22	99.50	98.40	1.09	4.45
Without FIG	97.08	97.45	97.37	97.24	1.41	3.49
Without YaTC	96.56	95.92	96.75	89.83	14.80	4.91
Without TrafficFormer	95.21	96.35	94.81	94.14	8.53	4.81

Note: Higher Known Macro-F1, AUROC, AUPRC, and Unknown F1 are better; lower UFAR and Known FRR are better.

DDoS achieves 99.45% AUPRC, but unknown samples account for 97.12% of its original test pool and UFAR remains 16.61%. Thus, under class imbalance, high AUPRC does not imply a low false acceptance rate at a fixed threshold. Rejection performance therefore requires joint assessment using ranking and threshold-dependent metrics. Together with VNAT and USTC, these results show effective overall unknown detection and recognition for most classes, while leaving room for improvement on unknown attacks whose feature distributions overlap those of known classes.

D. Rejection Mechanism and Component Ablations

1) Roles of Global and Local Empirical Support: To separate full-system gains from the contribution of the rejection rule, we fix the encoders, fused features $h(x)$, and classification outputs $\ell(x)$ across the four VNAT folds and change only the unknown-detection score. MSP and Energy use classification outputs. DES-v0 uses the squared distance to the nearest empirical centroid. DES-v1 additionally incorporates the mean Euclidean distance to the 10 nearest neighbors within the nearest-centroid class. Each score independently calibrates its threshold on the known-class validation set. Table 4-8 presents the results.

Local nearest-neighbor information further improves unknown detection on fixed representations. Compared with DES-v0, DES-v1 increases mean AUROC from 97.43% to 99.22%, reduces UFAR from 5.59% to 1.09%, and reduces Known FRR from 4.87% to 4.45%. AUROC improves in every fold, increasing from 94.47% to 98.76% for Remote Access and from 98.24% to 99.54% for Communication. This comparison directly supports the combination of global and local empirical support in Chapter 3: neighborhoods of actual samples provide rejection information beyond class means.

The File Transfer fold further illustrates the trade-off between the two error rates. DES-v1 reduces Known FRR from 5.03% to 4.61%, while UFAR increases from 0.36% to 0.56%. Local support therefore primarily improves ranking across the four folds and average acceptance behavior. A specific operating point still requires joint assessment of false rejection of known services and false acceptance of unknown traffic.

2) View Contributions: We ablate the FIG, YaTC, and TrafficFormer branches individually under the same four-fold VNAT configuration to assess each branch's contribution to the final representation. The encoders, Gaussian adapters, and normalization parameters remain frozen. All configurations retain the same fusion-head architecture and training settings for comparability. Table 4-9 reports closed-set classification and unknown-detection performance.

Removing any branch degrades some key detection metrics in Table 4-9. Removing YaTC decreases unknown-class F1 from 98.40% to 89.83% and increases UFAR from 1.09% to 14.80%, indicating the importance of multilevel byte representations for separating unknown traffic from known classes. Removing TrafficFormer decreases AUPRC from 99.50% to 94.81% and unknown-class F1 to 94.14%, indicating that contextual information in byte sequences complements known-class discrimination and unknown detection. Removing FIG decreases AUROC and AUPRC to 97.45% and 97.37%, respectively, decreases unknown-class F1 to 97.24%, and

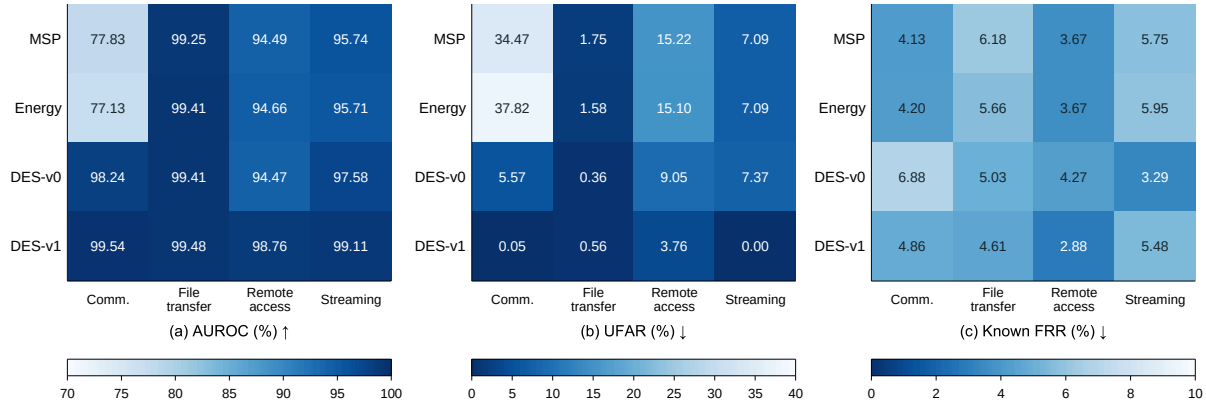


Fig. 4-3. Ablation of rejection scores on identical three-view representations. Columns denote the four unknown services, and rows denote the four scores. The figure reports AUROC, UFAR, and Known FRR in sequence; all cells are percentages.

TABLE 4-10
SENSITIVITY TO VALIDATION-QUANTILE THRESHOLDS ON VNAT (%, DES-V1, EQUAL-WEIGHT MEAN OVER FOUR FOLDS, ORIGINAL TEST-POOL PROPORTIONS)

Validation quantile	Unknown F1 ↑	UFAR ↓	Known FRR ↓
P90	97.26	0.91	10.02
P95	98.40	1.09	4.45
P99	85.94	20.98	1.23

increases UFAR to 1.41%. These results indicate that the flow interaction graph provides structural information for traffic discrimination and unknown detection by modeling relationships between packets. Overall, the three views provide complementary information that jointly improves known-class recognition and unknown-class rejection.

E. Sensitivity to the Rejection Threshold and Test Distribution

1) Effect of the Rejection Threshold: The preceding results use the fixed P95 operating point. We retain the VNAT network, training support bank, and score, and additionally use P90 and P99 of the known-class validation scores to examine false acceptance and false rejection. Table 4-10 uses the existing four-fold results, with P95 remaining the primary operating point.

The validation quantile directly controls the trade-off between acceptance of known samples and missed detection of unknown samples. P90 reduces mean UFAR to 0.91% but increases Known FRR to 10.02%. P99 reduces Known FRR to 1.23% but increases UFAR to 20.98%. A more permissive threshold increases known acceptance and may also increase missed unknown detections. All three operating points share the same score, so AUROC and AUPRC remain unchanged. Fixed-threshold error rates thus complement ranking metrics by characterizing acceptance behavior.

2) Unknown Proportion and Class Composition: Threshold sensitivity examines acceptance criteria under a fixed distribution, whereas mixed testing examines sample composition. From the frozen score pools for the ten CIC attacks, we draw 2,000 flows per trial and repeat the experiment using 20 deterministic seeds. With equal known and unknown proportions and equal representation of the ten unknown classes, DES-v1 achieves mean AUROC of 98.10%, unknown-class F1 of 93.12%, UFAR of 6.71%, and Known FRR of 7.07%. The model and P95 threshold remain frozen throughout.

Ranking performance and conditional error rates remain relatively stable as the unknown proportion changes. With the ten unknown classes internally balanced and the known:unknown ratio changing from 9:1 to 1:9, AUROC ranges from 98.04% to 98.13% and UFAR from 6.73% to 6.95%. AUPRC increases from 85.26% to 99.73% as the unknown proportion increases. In contrast, at a fixed 1:1 ratio, mixing only Bot and DDoS increases UFAR to 33.84%. Changes in the composition of normal traffic and PortScan also affect Known FRR. Together with the per-attack results, these findings indicate that the current detector is more sensitive to unknown-class composition than to the overall unknown proportion. Mixed-test averages should therefore accompany class-level error rates.

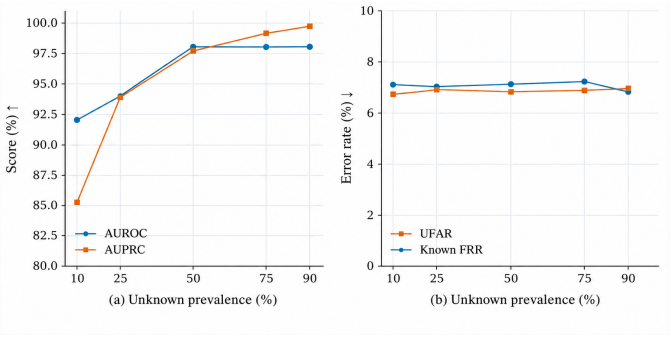


Fig. 4-4. Proportion and composition sensitivity on the frozen CIC score pools. (a) AUROC and AUPRC at different unknown proportions; (b) corresponding UFAR and Known FRR.

F. Discussion and Limitations

Limitations: We validate the proposed method on multiple public datasets and under different unknown-class holdout settings, but dataset provenance and traffic distributions constrain the experiments. The datasets differ in capture environments, application types, and traffic composition. Unknown-class

feature distributions also strongly affect rejection performance. The model still falsely accepts some unknown traffic whose features resemble those of known classes.

Future work: We will expand dataset and experimental coverage through cross-dataset and cross-network-environment generalization tests to assess adaptation to different traffic distributions. To address feature overlap between unknown and known classes, we will investigate more effective local feature modeling and rejection strategies for difficult unknown classes. We will also examine the effects of packet padding, segmentation, and traffic perturbations on multiview representations and explore detection methods that depend less on raw bytes to improve robustness in complex network environments.